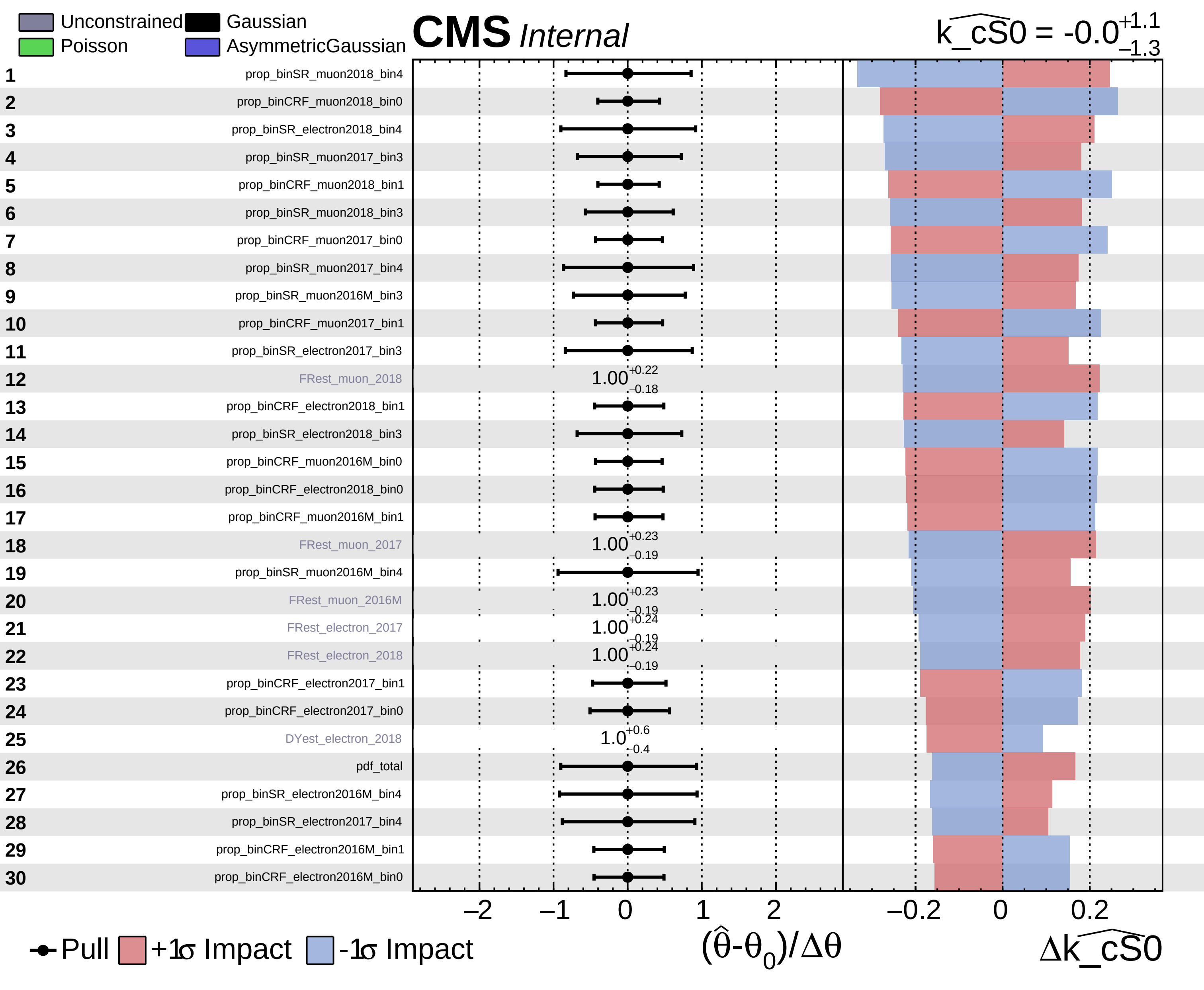
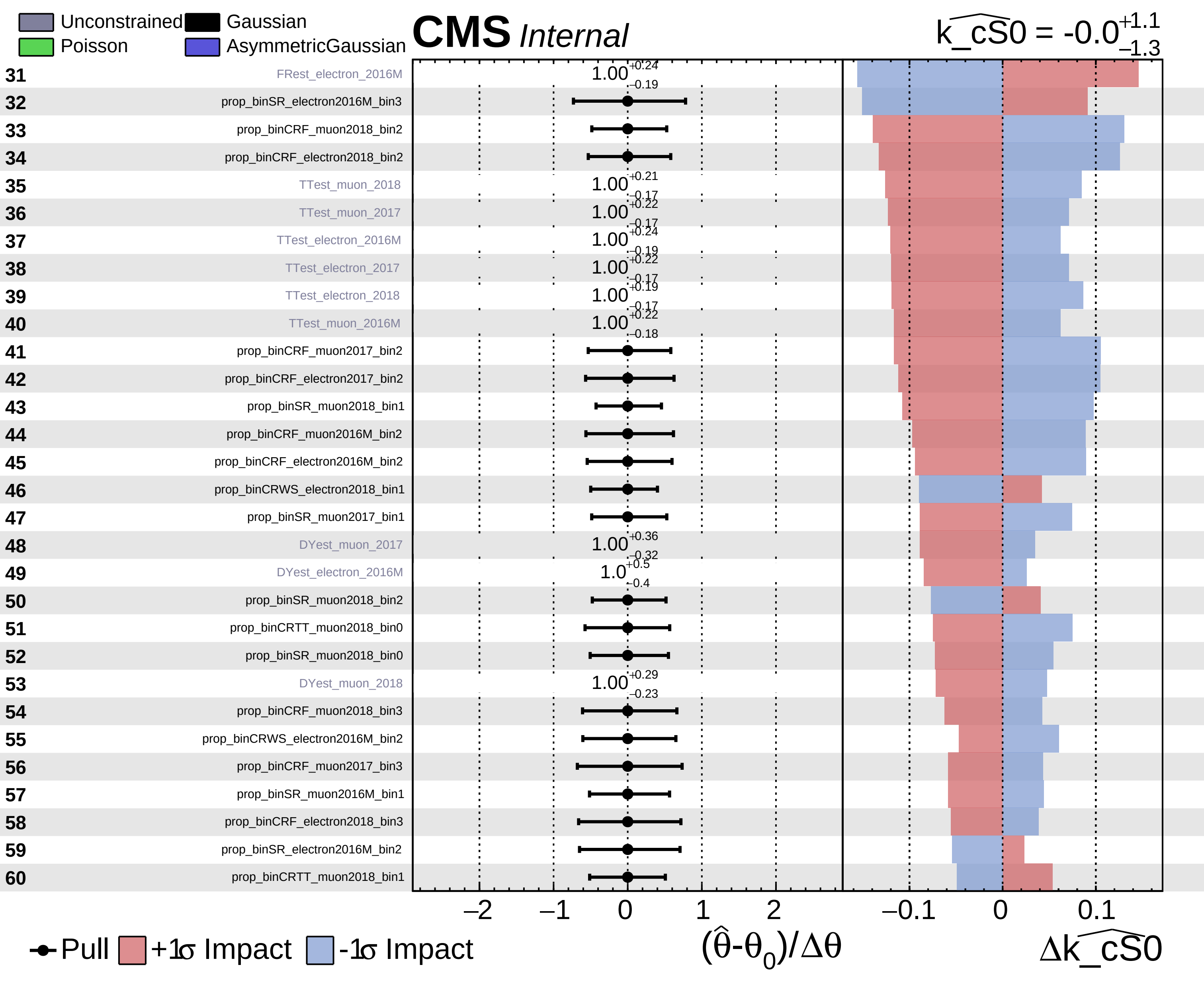
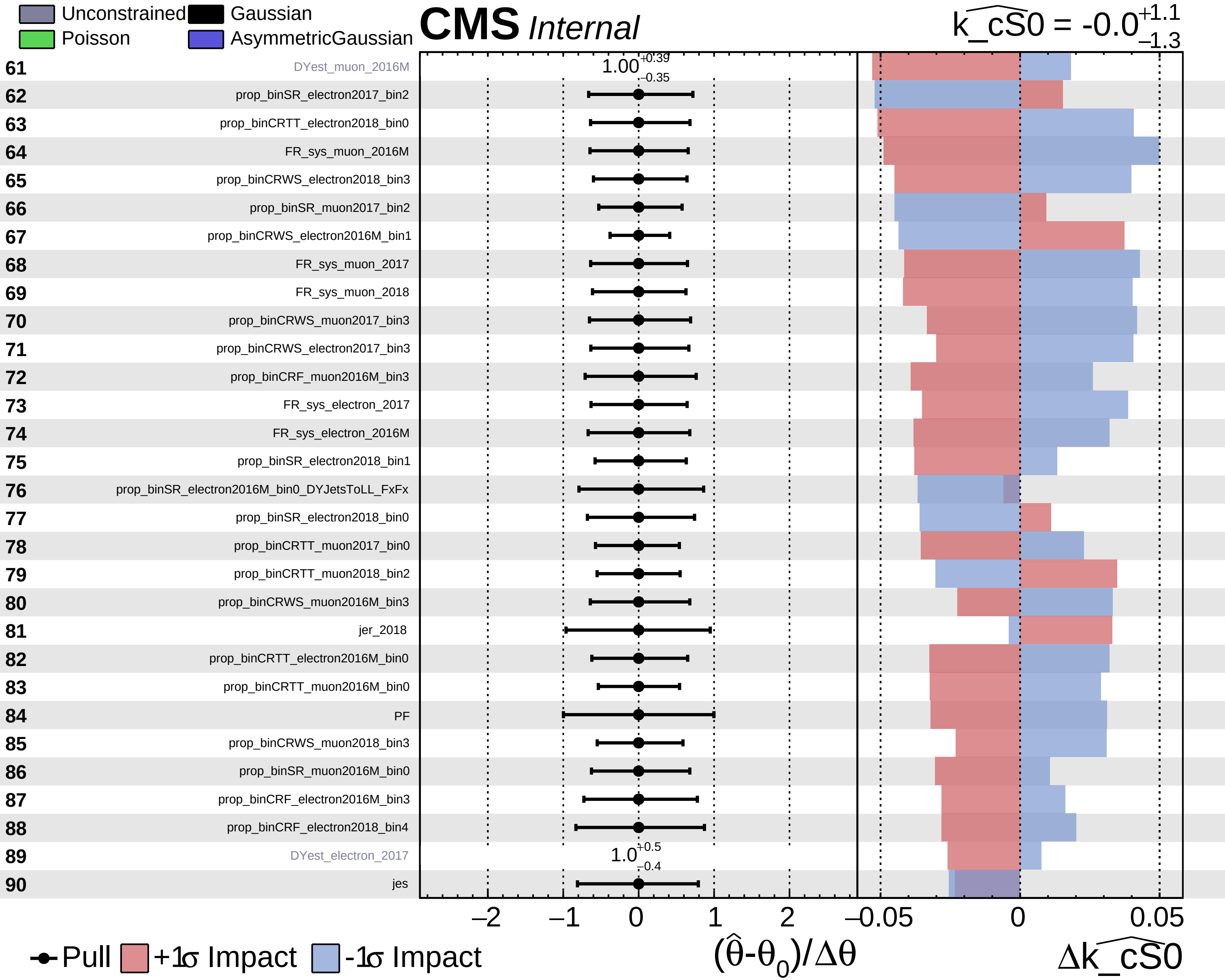


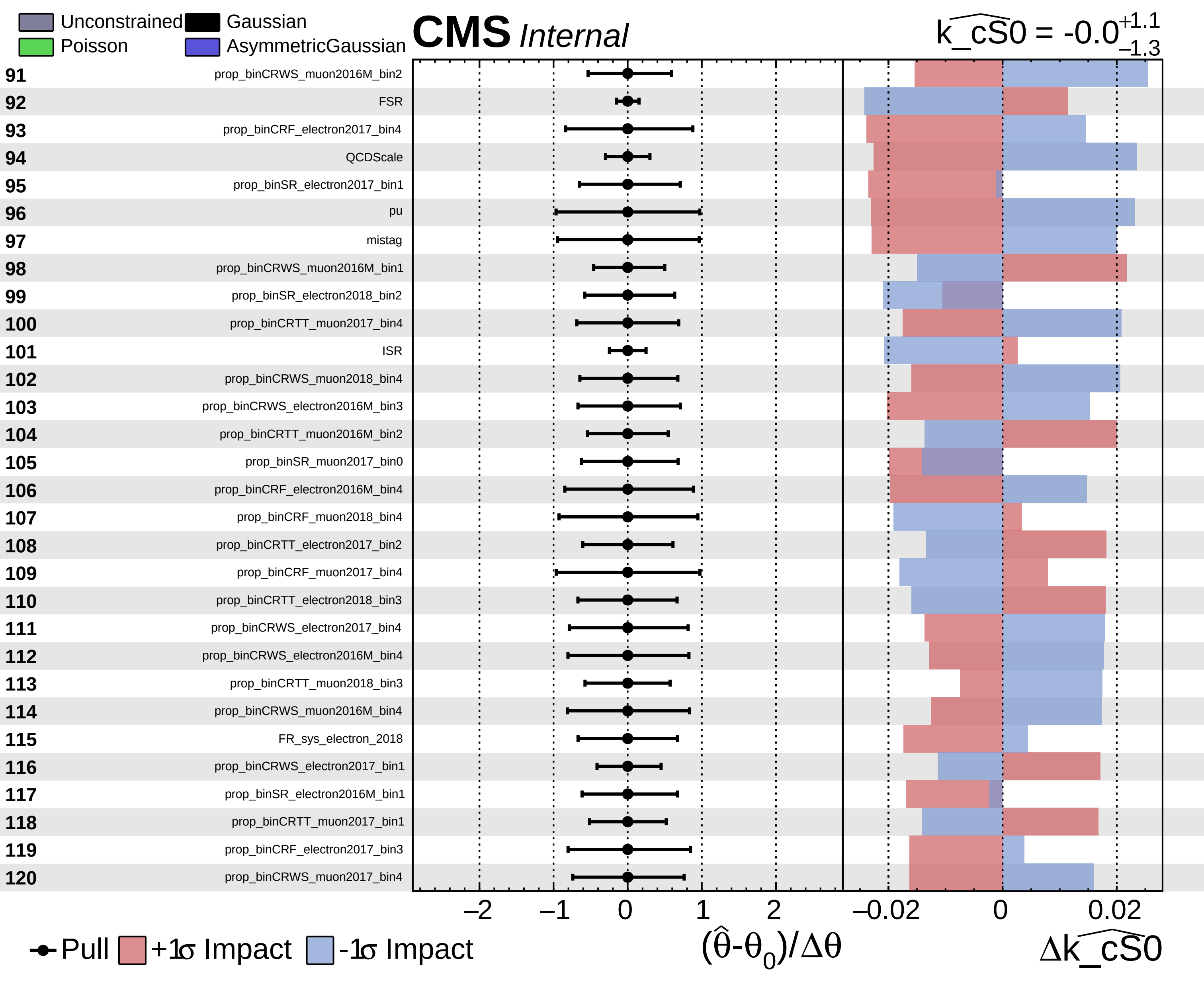
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS0} = -0.0^{+1.1}_{-1.3}$
1 prop_binSR_muon2018_bin4
2 prop_binCRF_muon2018_bin0
3 prop_binSR_electron2018_bin4
4 prop_binSR_muon2017_bin3
5 prop_binCRF_muon2018_bin1
6 prop_binSR_muon2018_bin3
7 prop_binCRF_muon2017_bin0
8 prop_binSR_muon2017_bin4
9 prop_binSR_muon2016M_bin3
10 prop_binCRF_muon2017_bin1
11 prop_binSR_electron2017_bin3
12 FRest_muon_2018 $1.00^{+0.22}_{-0.18}$
13 prop_binCRF_electron2018_bin1
14 prop_binSR_electron2018_bin3
15 prop_binCRF_muon2016M_bin0
16 prop_binCRF_electron2018_bin0
17 prop_binCRF_muon2016M_bin1
18 FRest_muon_2017 $1.00^{+0.23}_{-0.19}$
19 prop_binSR_muon2016M_bin4
20 FRest_muon_2016M $1.00^{+0.23}_{-0.19}$
21 FRest_electron_2017 $1.00^{+0.24}_{-0.19}$
22 FRest_electron_2018 $1.00^{+0.24}_{-0.19}$
23 prop_binCRF_electron2017_bin1
24 prop_binCRF_electron2017_bin0
25 DYest_electron_2018 $1.0^{+0.6}_{-0.4}$
26 pdf_total
27 prop_binSR_electron2016M_bin4
28 prop_binSR_electron2017_bin4
29 prop_binCRF_electron2016M_bin1
30 prop_binCRF_electron2016M_bin0
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS0}$



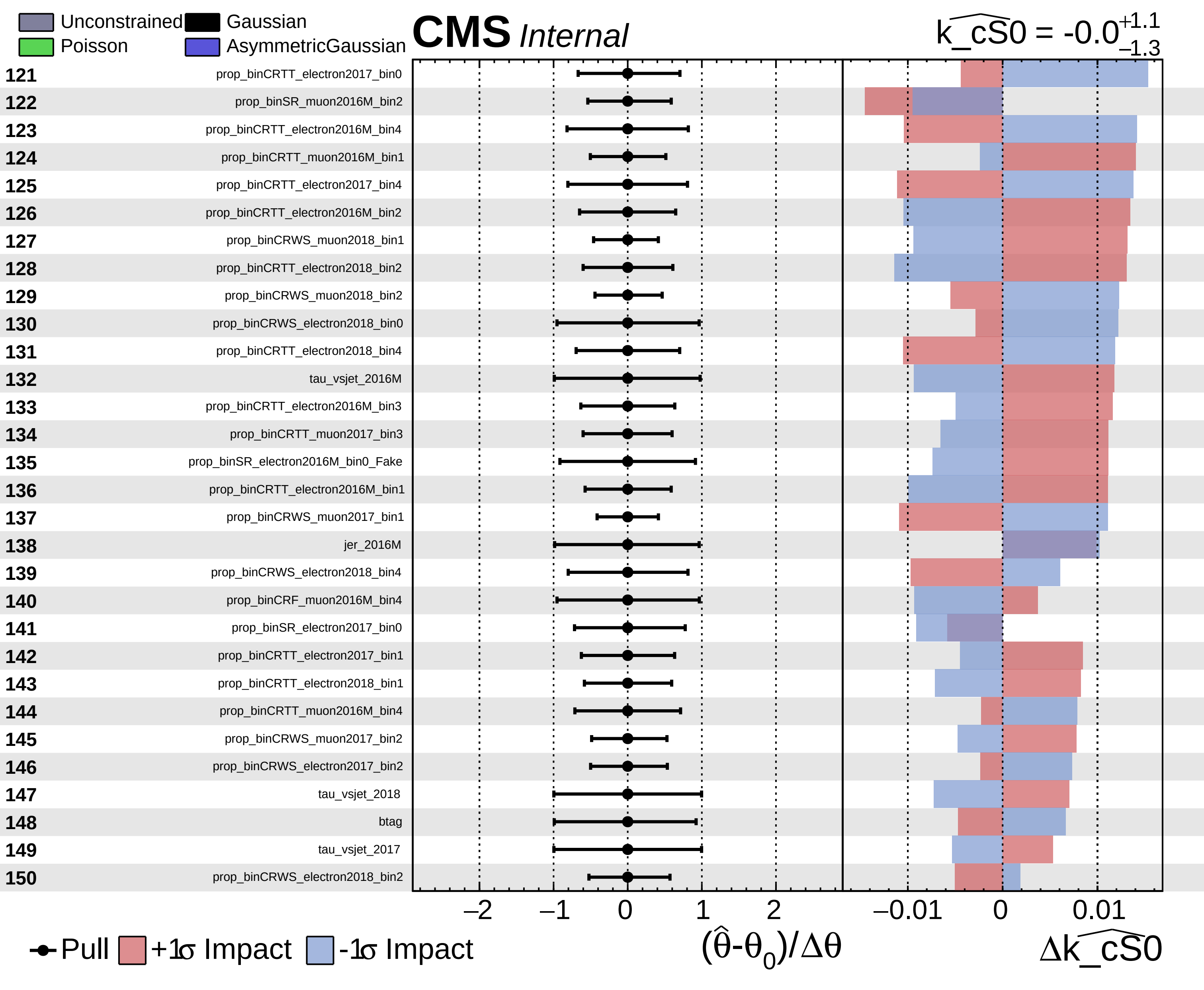
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS0 = -0.0 +1.1 −1.3
31 FRest_electron_2016M 1.00 +0.24 −0.19
32 prop_binSR_electron2016M_bin3
33 prop_binCRF_muon2018_bin2
34 prop_binCRF_electron2018_bin2
35 TTest_muon_2018 1.00 +0.21 −0.17
36 TTest_muon_2017 1.00 +0.22 −0.17
37 TTest_electron_2016M 1.00 +0.24 −0.19
38 TTest_electron_2017 1.00 +0.22 −0.17
39 TTest_electron_2018 1.00 +0.19 −0.17
40 TTest_muon_2016M 1.00 +0.22 −0.18
41 prop_binCRF_muon2017_bin2
42 prop_binCRF_electron2017_bin2
43 prop_binSR_muon2018_bin1
44 prop_binCRF_muon2016M_bin2
45 prop_binCRF_electron2016M_bin2
46 prop_binCRWS_electron2018_bin1
47 prop_binSR_muon2017_bin1
48 DYest_muon_2017 1.00 +0.36 −0.32
49 DYest_electron_2016M 1.0 +0.5 −0.4
50 prop_binSR_muon2018_bin2
51 prop_binCRTT_muon2018_bin0
52 prop_binSR_muon2018_bin0
53 DYest_muon_2018 1.00 +0.29 −0.23
54 prop_binCRF_muon2018_bin3
55 prop_binCRWS_electron2016M_bin2
56 prop_binCRF_muon2017_bin3
57 prop_binSR_muon2016M_bin1
58 prop_binCRF_electron2018_bin3
59 prop_binSR_electron2016M_bin2
60 prop_binCRTT_muon2018_bin1
−2 −1 0 1 2
−0.1 0 0.1
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cS0



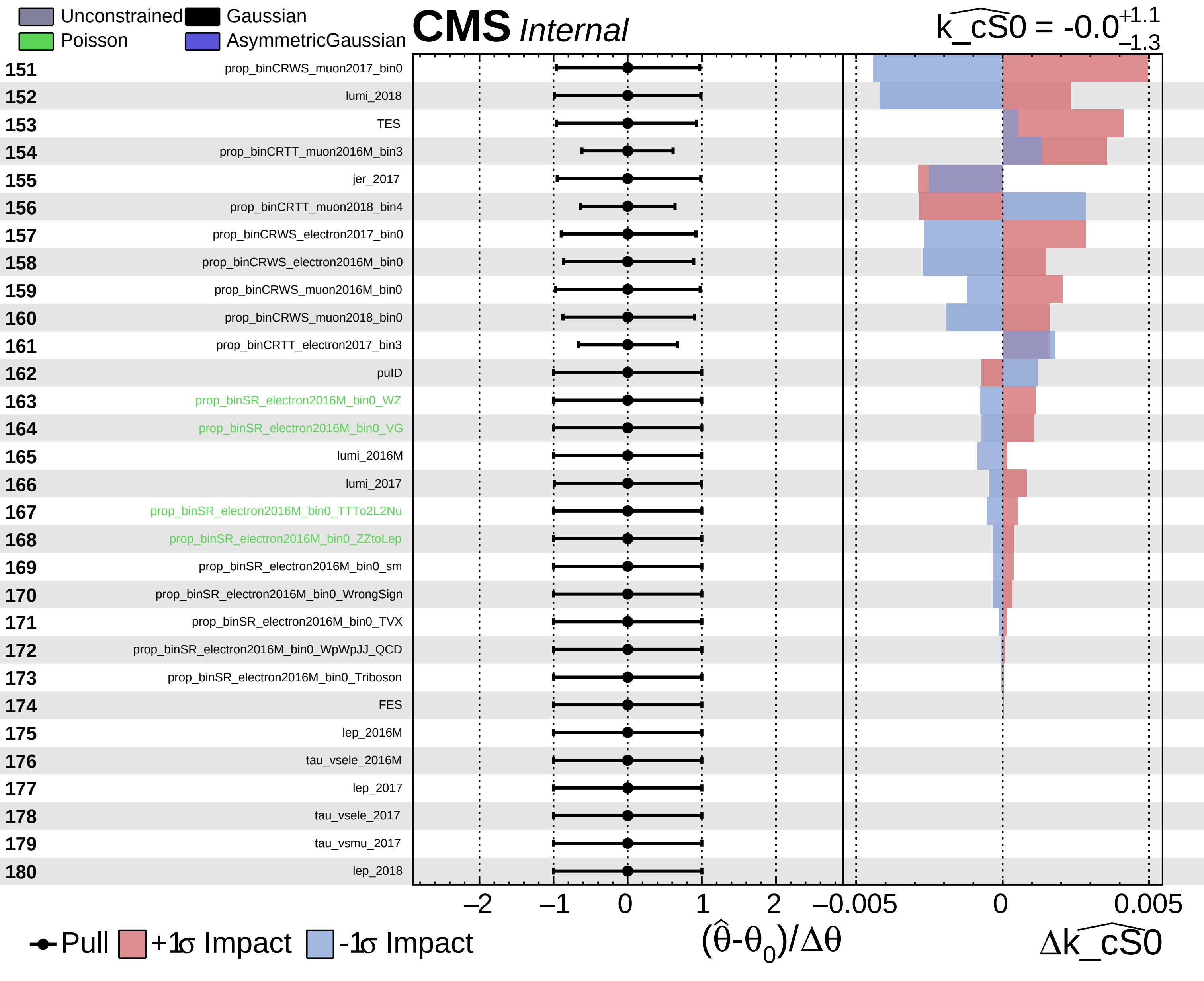
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS0 = -0.0 +1.1 −1.3
61 DYest_muon_2016M 1.00 +0.39 −0.35
62 prop_binSR_electron2017_bin2
63 prop_binCRTT_electron2018_bin0
64 FR_sys_muon_2016M
65 prop_binCRWS_electron2018_bin3
66 prop_binSR_muon2017_bin2
67 prop_binCRWS_electron2016M_bin1
68 FR_sys_muon_2017
69 FR_sys_muon_2018
70 prop_binCRWS_muon2017_bin3
71 prop_binCRWS_electron2017_bin3
72 prop_binCRF_muon2016M_bin3
73 FR_sys_electron_2017
74 FR_sys_electron_2016M
75 prop_binSR_electron2018_bin1
76 prop_binSR_electron2016M_bin0_DYJetsToLL_FxFx
77 prop_binSR_electron2018_bin0
78 prop_binCRTT_muon2017_bin0
79 prop_binCRTT_muon2018_bin2
80 prop_binCRWS_muon2016M_bin3
81 jer_2018
82 prop_binCRTT_electron2016M_bin0
83 prop_binCRTT_muon2016M_bin0
84 PF
85 prop_binCRWS_muon2018_bin3
86 prop_binSR_muon2016M_bin0
87 prop_binCRF_electron2016M_bin3
88 prop_binCRF_electron2018_bin4
89 DYest_electron_2017 1.0 +0.5 −0.4
90 jes
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS0}$



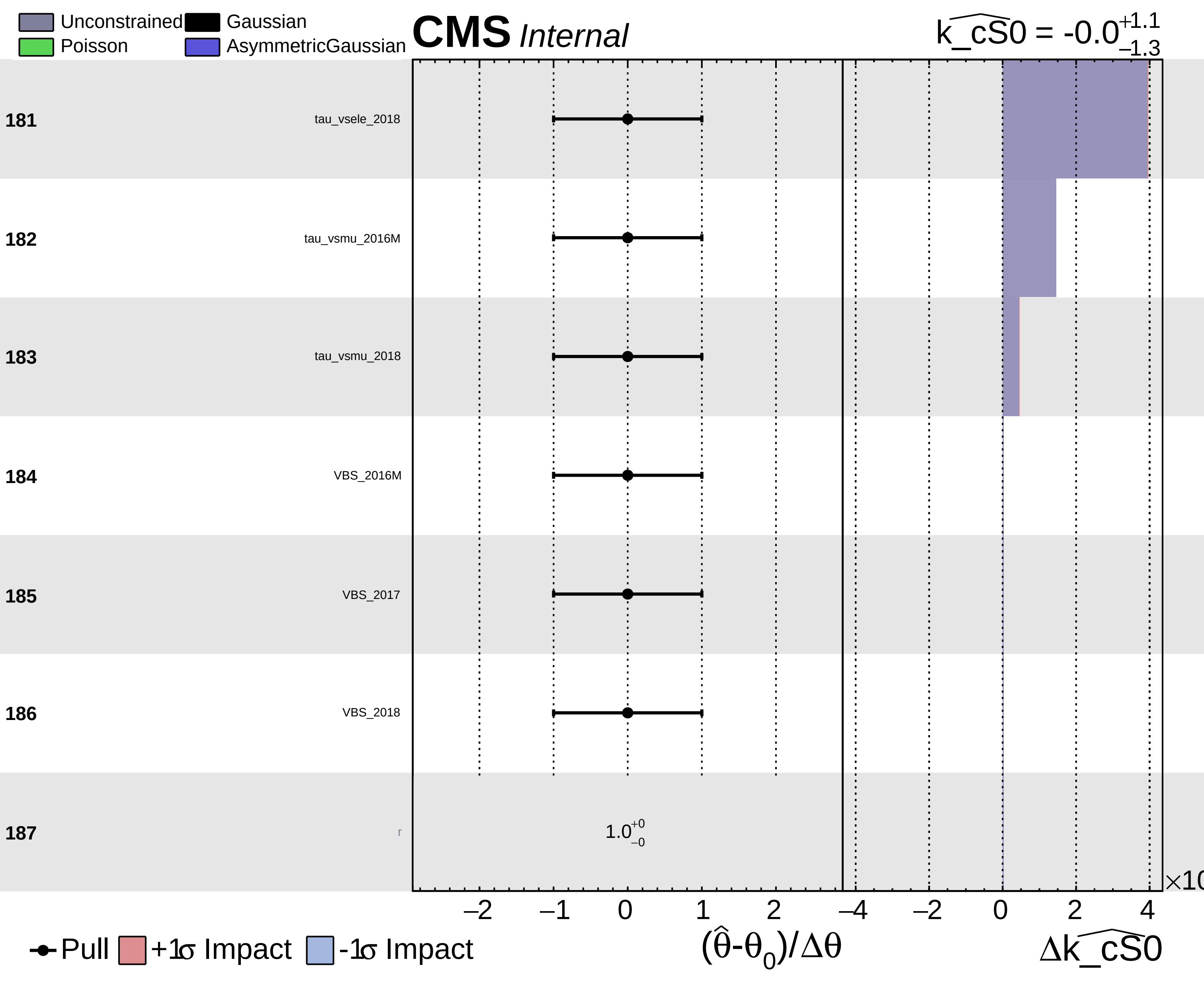
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS0} = -0.0^{+1.1}_{-1.3}$
91 prop_binCRWS_muon2016M_bin2
92 FSR
93 prop_binCRF_electron2017_bin4
94 QCDScale
95 prop_binSR_electron2017_bin1
96 pu
97 mistag
98 prop_binCRWS_muon2016M_bin1
99 prop_binSR_electron2018_bin2
100 prop_binCRTT_muon2017_bin4
101 ISR
102 prop_binCRWS_muon2018_bin4
103 prop_binCRWS_electron2016M_bin3
104 prop_binCRTT_muon2016M_bin2
105 prop_binSR_muon2017_bin0
106 prop_binCRF_electron2016M_bin4
107 prop_binCRF_muon2018_bin4
108 prop_binCRTT_electron2017_bin2
109 prop_binCRF_muon2017_bin4
110 prop_binCRTT_electron2018_bin3
111 prop_binCRWS_electron2017_bin4
112 prop_binCRWS_electron2016M_bin4
113 prop_binCRTT_muon2018_bin3
114 prop_binCRWS_muon2016M_bin4
115 FR_sys_electron_2018
116 prop_binCRWS_electron2017_bin1
117 prop_binSR_electron2016M_bin1
118 prop_binCRTT_muon2017_bin1
119 prop_binCRF_electron2017_bin3
120 prop_binCRWS_muon2017_bin4
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS0}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS0 = -0.0 +1.1 −1.3
121 prop_binCRTT_electron2017_bin0
122 prop_binSR_muon2016M_bin2
123 prop_binCRTT_electron2016M_bin4
124 prop_binCRTT_muon2016M_bin1
125 prop_binCRTT_electron2017_bin4
126 prop_binCRTT_electron2016M_bin2
127 prop_binCRWS_muon2018_bin1
128 prop_binCRTT_electron2018_bin2
129 prop_binCRWS_muon2018_bin2
130 prop_binCRWS_electron2018_bin0
131 prop_binCRTT_electron2018_bin4
132 tau_vsjet_2016M
133 prop_binCRTT_electron2016M_bin3
134 prop_binCRTT_muon2017_bin3
135 prop_binSR_electron2016M_bin0_Fake
136 prop_binCRTT_electron2016M_bin1
137 prop_binCRWS_muon2017_bin1
138 jer_2016M
139 prop_binCRWS_electron2018_bin4
140 prop_binCRF_muon2016M_bin4
141 prop_binSR_electron2017_bin0
142 prop_binCRTT_electron2017_bin1
143 prop_binCRTT_electron2018_bin1
144 prop_binCRTT_muon2016M_bin4
145 prop_binCRWS_muon2017_bin2
146 prop_binCRWS_electron2017_bin2
147 tau_vsjet_2018
148 btag
149 tau_vsjet_2017
150 prop_binCRWS_electron2018_bin2
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS0



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS0} = -0.0^{+1.1}_{-1.3}$
151 prop_binCRWS_muon2017_bin0
152 lumi_2018
153 TES
154 prop_binCRTT_muon2016M_bin3
155 jer_2017
156 prop_binCRTT_muon2018_bin4
157 prop_binCRWS_electron2017_bin0
158 prop_binCRWS_electron2016M_bin0
159 prop_binCRWS_muon2016M_bin0
160 prop_binCRWS_muon2018_bin0
161 prop_binCRTT_electron2017_bin3
162 puID
163 prop_binSR_electron2016M_bin0_WZ
164 prop_binSR_electron2016M_bin0_VG
165 lumi_2016M
166 lumi_2017
167 prop_binSR_electron2016M_bin0_TTTo2L2Nu
168 prop_binSR_electron2016M_bin0_ZZtoLep
169 prop_binSR_electron2016M_bin0_sm
170 prop_binSR_electron2016M_bin0_WrongSign
171 prop_binSR_electron2016M_bin0_TVX
172 prop_binSR_electron2016M_bin0_WpWpJJ_QCD
173 prop_binSR_electron2016M_bin0_Triboson
174 FES
175 lep_2016M
176 tau_vsele_2016M
177 lep_2017
178 tau_vsele_2017
179 tau_vsmu_2017
180 lep_2018
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS0}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS0 = -0.0+1.1 −1.3
181
tau_vsele_2018
182
tau_vsmu_2016M
183
tau_vsmu_2018
184
VBS_2016M
185
VBS_2017
186
VBS_2018
187
r
1.0+0 −0
−2 −1 0 1 2
−4 −2 0 2 4
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS0